

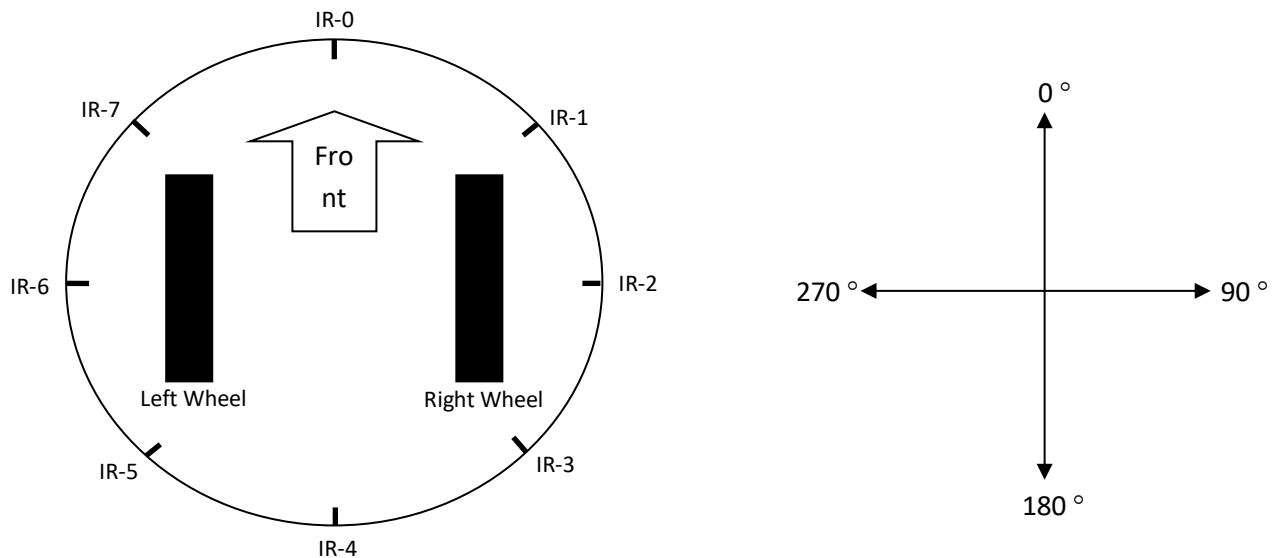
AML-Assignment#1 - RC

(Simple Reactive/Functional Control (PyRCSimbot))

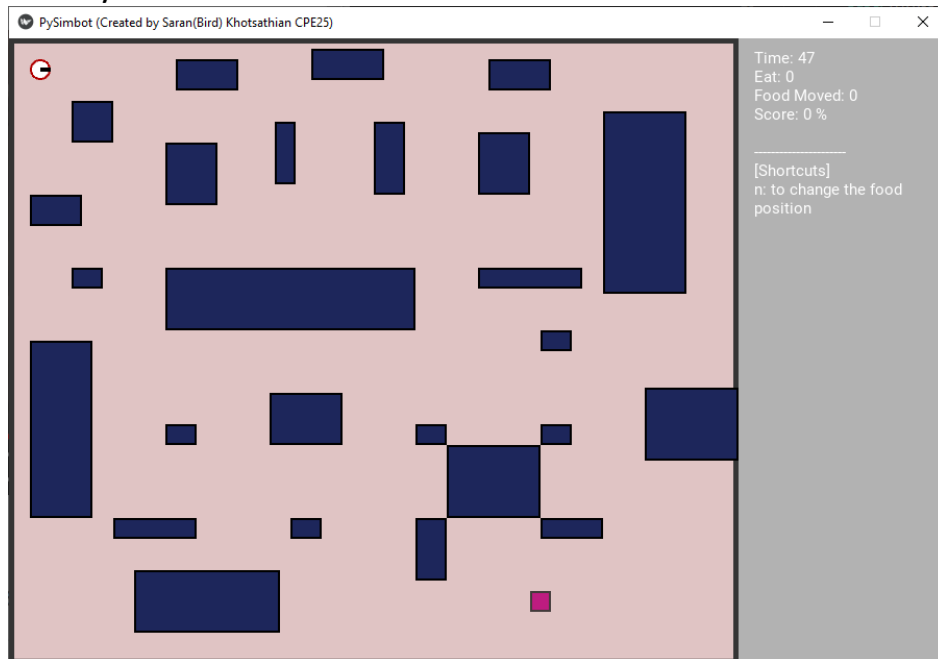


Let's design a reactive control for **a very simple simulated robot (PyRCSimbot)**: you can download from the class website). In this assignment, we will give you simulation robot codes (in python, of course), which you will work on it. There are some limitations and specifications about the robot we are using, as follows...

- The robot has only 8 infrared sensors (IR-0, IR-1, IR-2, ..., IR-7) locating around the robot. Each sensor reads the distance to the nearest object in its direction. The possible value is 0 to 100 pixels. It means the sensors are only checked for the 100 pixels ahead in its direction.
 - The sensory information can be gathered by calling **self.distance()**. For example **IR = self.distance()**. Then, the updated sensory information are in a array variable of **IR[8]**. The index is used relating to sensor's position on the robot. For example, we use **IR[0]** keeps the information of sensor IR-0.
- The direction of the robot is referred to its head position which is referred to 0 degree. The other directions are then referred in clockwise direction.
 - In addition, the robot can smell for direction of food. By calling **self.smell()**, the robot can get the direction of food referred to the heading position of the robot.
 - For the actions, there are 3 commands available to use. There are **self.move(distance)**, **self.turn(angle)**. Please use only these commands.
 - Note :
 - positive angle is clockwise
 - default distance = 1
 - distance should not more than 10
 - The robot program should be written in a file named **run.py**. You can do or modify whatever you want to program it moving.
 - The environment is shown below. The robot is a circle with a small circle on it. The small circle indicates the heading position of the robot. There are 26 obstacle boxes on the field. The robot cannot move pass it. If the robot is forced to move into the obstacle it will force to return the previous position. The green square on the right is the food and it can be smelled by the robot. The robot will get a direction (reference to the direction of its heading) to the food when it tries to smell.



- The environment is static. All obstacles are not moving during execution. The food is also not moving. The robot start position and head direction are the same every execution.



- Your task is

Writing a reactive control program with limited capacities; so that the robot has to move pass the obstacles toward the food as fast as it can.

- When we talking about reactive control, we are talking about the system that uses the current input information to select the actions without looking ahead or planning.

Please do it wisely and the most importance thing is please do it by yourself.

Don't use any random function. Let's have fun with it. Don't worry about the score toward your robot performance. I love to see you efforts in solving the problem the most. At the end, you will learn a lot. 😊

HINT ...

Define safety distance 30
Define closed distance 5
Define hit distance 0

If we use only 3 front IR sensors, say

- IR[0] is at front,
- IR[1] is at right side pointing to front right,
- and IR[7] is at left side pointing to front left

There are 4 conditions that we have to control for different movements.

- 1) If all front three IR sensors are clear for safety distance to any objects, then make the robot
 - o move forward, and
 - o turn toward to the food direction (for how many degree is your decision)
- 2) If the front IR sensor (IR[0]) is clear for safety distance to any objects and the other two sensors (IR[1] and IR[7]) are ok for the closed distance, then make the robot
 - o only move forward
- 3) If the front IR sensor (IR[0]) is clear for safety distance to any objects and the other two sensors (IR[1] or IR[7]) are too close to the objects, less than the closed distance, then make the robot
 - o turn to direction that far from the objects (say, if IR[1] is less than IR[7], mean that IR[1] (that is on the right side) is more closed to the object than IR[7] (that is on the left side), then the robot should turn left, ... Likewise, if IR[7] is less than IR[1], the robot should turn right)
- 4) If the front IR sensors (IR[0]) is in the safety distance to hit any object, then make the robot
 - o turn toward one direction (for which direction and how many degree are your decisions)

Restrictions

- 1. Don't use any randomness in your code.**
- 2. Clip should show at least your robot could reach the food once by your code.**

Submissions

1. Post your assignment's running clip to the FB group. Write all students name and ID, with the title, for example ... "RC assignment"
2. Use this form (<https://forms.cloud.microsoft/r/2H45whtJev>) for submission.
3. Please submit before deadline, if you submit late, your score will be deducted.